The Small-Argument Asymptotic Expansion of the Fabius Function

A general formula for every coefficient

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Abstract

As $x \to 0^+$ the Fabius function F is flat to infinite order, so it has no Taylor expansion at 0; the correct description is an exponentially small asymptotic expansion in the lower-Lambert phase $\lambda(x)$, whose coefficients A_j are one-periodic functions of λ rather than constants. Until now those coefficients existed only as the output of a chain of recursions — a differential recursion for the saddle jets, a formal exponential recursion, a Gaussian contraction, and a formal logarithm — and only A_1 had ever been written down. This note closes the chain. We solve the differential recursion in closed form using signed Stirling numbers of the first kind and harmonic numbers, collapse the exponent coefficients to a single power of i times a jet term plus a universal jet-free tail, and assemble a completely explicit, non-recursive formula for every A_j as a finite sum over compositions, subsets and Gaussian moments. As applications we compute A_2 and A_3 explicitly for the first time, identify the highest-derivative term of A_j as $(-1)^j \Psi^{(2j)}/(2^j j!\, L^{2j})$, deduce the Gamma–zeta Fourier form and the amplitude law of every coefficient, and verify the whole chain numerically against exact rational values of $F(2^{-n})$.

Contents

1 Introduction

Throughout, F is the canonical bounded Fabius function on $[0,1]$: the cumulative distribution function of

$$X = \sum_{k\geq 1} 2^{-k} U_k, \qquad U_1, U_2, \ldots \text{ independent uniform on } [0,1].$$

It is smooth, vanishes on $(-\infty, 0]$, equals 1 on $[1,\infty)$, satisfies $F(x) + F(1-x) = 1$, and obeys $F'(x) = 2F(2x)$ on $[0, \frac{1}{2}]$. Every derivative vanishes at the origin, so F is flat there and no Taylor expansion exists.

This note develops the behaviour of $F(x)$ as $x \to 0^+$ in detail, and in particular gives a general formula for every coefficient of its all-orders expansion. It is a companion to the formal development in `Analysis/FabiusFunction` and to the article in `docs/Fabius_Function_and_Rvachev_Up/`, and it is self-contained: everything needed to compute any coefficient by hand is stated here.

1.1 What was already known, and what is new

The pre-existing development already contained the corrected sharp main term $\mathcal{M}(x)$ with its genuine nonconstant periodic correction, a proof that deleting that correction makes the standard printed formula false, and an all-orders expansion

$$\log F(x) = \mathcal{M}(x) + \sum_{j\geq 1} \frac{A_j(\lambda)}{\lambda^j}$$

in which the coefficient functions A_j were *defined only by a recursion.* Only A_1 was ever displayed.

What this note adds:

1. A closed form for the saddle jets (section 4). The differential recursion is solved: the n-th jet is a harmonic-number constant plus a Stirling-weighted combination of the first $n+1$ derivatives of the periodic correction. This removes the only non-algebraic recursion from the chain.
2. A closed form for the exponent coefficients (section 5), with the factorials and the spurious extra power of i cancelled.
3. A fully explicit, non-recursive formula for every coefficient (section 6): A_j is a finite triple sum over compositions, subsets and Gaussian moments.
4. The explicit coefficients A_2 and A_3 (section 7), which did not exist anywhere before, in two equivalent forms.
5. The leading-derivative law (section 8): the top term of A_j is $(-1)^j\Psi^{(2j)}(u)/(2^j j!\,L^{2j})$, which explains why the oscillation of the coefficients grows with their order.
6. The Gamma–zeta Fourier form of every coefficient and the resulting amplitude law (section 9).
7. An independent end-to-end numerical verification (section 10) against exact rational values of $F(2^{-n})$.

1.2 Notation

symbol	meaning
L	$\log 2 = 0.69314718055994530942\ldots$
γ	Euler–Mascheroni constant
γ_1	first Stieltjes constant, $\zeta(1+s) = 1/s + \gamma - \gamma_1 s + \mathcal{O}(s^2)$
H_n	n-th harmonic number, $H_0 = 0$
Ψ	the centered one-periodic correction of section 3
$\Psi^{(m)}$	m-th derivative of Ψ
$\lambda = \lambda(x)$	the lower-Lambert phase of section 2
$\mathcal{M}(x)$	the sharp main term of section 3
A_j	the j-th expansion coefficient, a one-periodic function
a_j	the j-th *mass* coefficient, before taking the logarithm
d_n	the n-th bounded exponent jet of section 4
p_n	the n-th periodic jet of section 4
s_n	$(-1)^{n+1}n!$, the jet slope
$c(n,m)$	$[z^m]\prod_{k=1}^{n}(z-k)$, a signed Stirling number of the first kind
$(N-1)!!$	$1\cdot 3\cdot 5\cdots(N-1)$, the N-th standard Gaussian moment for even N

Z denotes a standard Gaussian variable, so $\mathbb{E}[Z^N] = (N-1)!!$ for even N and 0 for odd N.

2 The lower-Lambert phase

The saddle point of the Laplace inversion sits at radius $r = 2^\lambda$, where λ solves

$$\lambda\, 2^{-\lambda} = x, \qquad \text{equivalently} \qquad L\lambda = -\log x + \log\lambda. \tag{1}$$

This is the lower branch of the Lambert W function, $\lambda(x) = -W_{-1}(-Lx)/L$, defined and strictly decreasing for $0 < Lx < \mathrm{e}^{-1}$, with $\lambda(x) \to +\infty$ as $x \to 0^+$.

The expansion below is an expansion in $1/\lambda$, *not* in $1/(-\log x)$. The two agree to leading order, since $L\lambda = -\log x + \log(-\log x) - \log L + o(1)$, but they differ at every order after that; section 12 explains why substituting an expansion of λ into the coefficients is not a legitimate step.

For dyadic arguments $x = 2^{-n}$, equation equation (1) reads $L\lambda = nL + \log\lambda$, whose solution is conveniently obtained by the iteration $\lambda \mapsto n + \log(\lambda)/L$.

3 The periodic correction and the sharp main term

Let

$$G(-s) = \prod_{j\geq 1} \frac{1-\mathrm{e}^{-s/2^j}}{s/2^j}, \qquad \Lambda(s) = \log G(-s),$$

be the logarithm of the two-sided Laplace transform of the Fabius density. Removing the quadratic drift and the forward tail $T(s) = \sum_{m\geq 0} \log(1-\mathrm{e}^{-s2^m})$ produces the *exactly* one-periodic function

$$P(t) = \Lambda(2^t) + \frac{L}{2}\big(t^2 - t\big) + T(2^t), \qquad P(t+1) = P(t),$$

so that

$$\Lambda(s) = -\frac{(\log s)^2}{2L} + \frac{\log s}{2} + P(\log_2 s) + E(s), \qquad E = -T, \qquad 0 \leq E(s) \leq \frac{\mathrm{e}^{-s}}{(1-\mathrm{e}^{-s})^2}.$$

Write $\Psi(t) = P(t) - \bar{P}$ with $\bar{P} = \int_0^1 P$. Then Ψ is one-periodic, C^∞, of mean zero, and — this is the decisive point — *not constant.* Its Fourier coefficients are

$$\widehat{\Psi}(0) = 0, \qquad \widehat{\Psi}(k) = -\frac{\Gamma(-\chi_k)\,\zeta(1-\chi_k)}{L}, \qquad \chi_k = \frac{2\pi\mathrm{i}k}{L}, \quad k \neq 0, \tag{2}$$

which are nonzero for every $k \neq 0$ because Γ has no zeros and ζ has none on the line $\operatorname{Re} s = 1$. Numerically

$$\begin{aligned}
\widehat{\Psi}(1) &= (\ 7.55864754098 - 7.47010355898\,\mathrm{i}\ ) \times 10^{-7}, & |\widehat{\Psi}(1)| &= 1.06271162519 \times 10^{-6},\\
\widehat{\Psi}(2) &= (\ 3.24812347723 + 5.85175898712\,\mathrm{i}\ ) \times 10^{-13}, & |\widehat{\Psi}(2)| &= 6.69278636792 \times 10^{-13},\\
\widehat{\Psi}(3) &= (-2.81236523199 - 2.58674397916\,\mathrm{i}\ ) \times 10^{-19}, & |\widehat{\Psi}(3)| &= 3.82107872359 \times 10^{-19},
\end{aligned}$$

so Ψ itself oscillates by only about 2×10^{-6}. Section 13 records the numerical trap that makes these values easy to get wrong.

With $A_0^{\mathrm{c}} = (6\gamma^2 + 12\gamma_1 - \pi^2)/12$ and

$$C_F = \frac{A_0^{\mathrm{c}}}{L} - \frac{7L}{12} - \frac{\log\pi}{2} = -2.02798389863885942397 1\ldots,$$

the sharp main term is

$$\mathcal{M}(x) = -\frac{L}{2}\lambda^2 + \Big(1 + \frac{L}{2}\Big)\lambda - \frac{1}{2}\log\lambda + C_F + \Psi(\lambda), \tag{3}$$

and $\log F(x) = \mathcal{M}(x) + \mathcal{O}(1/\lambda)$. Deleting $\Psi(\lambda)$ invalidates that error term: the ratio $F(x)/\exp(\mathcal{M}(x) - \Psi(\lambda))$ has at least two distinct subsequential limits.

4 The saddle jets and their closed form

Let $q = \log G(-\cdot\,)$ evaluated on the saddle ray $r = 2^t$. Up to the exponentially small forward tail, the $(n+1)$-st scaled derivative has the form

$$r^{n+1} q^{(n+1)}(r) = s_n t + p_n(t), \qquad s_n = (-1)^{n+1} n!,$$

where the periodic jets p_n are determined by the differential recursion

$$p_0(t) = \frac{1}{2} + \frac{\Psi'(t)}{L}, \qquad p_{n+1}(t) = \frac{p_n'(t)}{L} - (n+1)p_n(t) + \frac{s_n}{L}. \tag{4}$$

The quantity that actually enters the saddle exponent is the *bounded exponent jet* $d_n = p_n + s_n$.

4.1 Solving the recursion

Write D for $\mathrm{d}/\mathrm{d}t$. The recursion equation (4) reads $p_{n+1} = \big(D/L - (n+1)\big)p_n + s_n/L$, and the operators $D/L - j$ commute, since

$$\Big(\frac{D}{L} - j\Big)\Big(\frac{D}{L} - k\Big) = \frac{D^2}{L^2} - (j+k)\frac{D}{L} + jk$$

is symmetric in j and k. Hence the homogeneous part of the solution is the falling-factorial operator $\prod_{k=1}^{n}(D/L - k)$ applied to p_0. Each inhomogeneous constant s_j/L is carried forward by $\prod_{k=j+2}^{n}(D/L - k)$, and a constant is an eigenvector of $D/L - k$ with eigenvalue $-k$, so its contribution is

$$\frac{s_j}{L}\,(-1)^{\,n-j-1}\frac{n!}{(j+1)!} = \frac{(-1)^n\, n!}{(j+1)L}.$$

Summing over $j = 0, \ldots, n-1$ produces the harmonic number. Finally $\prod_{k=1}^{n}(z-k) = \sum_{m=0}^{n} c(n,m) z^m$, so applying the operator polynomial to $p_0 = \frac{1}{2} + \Psi'/L$ gives the result.

4.2 The closed forms

Theorem 4.1 (Closed-form jets)**.** *For every $n \geq 0$ and every t,*

$$p_n(t) = (-1)^n n!\Big(\frac{1}{2} + \frac{H_n}{L}\Big) + \sum_{m=0}^{n} c(n,m)\,\frac{\Psi^{(m+1)}(t)}{L^{m+1}},$$
$$d_n(t) = (-1)^n n!\Big(\frac{H_n}{L} - \frac{1}{2}\Big) + \sum_{m=0}^{n} c(n,m)\,\frac{\Psi^{(m+1)}(t)}{L^{m+1}},$$

where $c(n,m) = [z^m]\prod_{k=1}^{n}(z-k)$ is a signed Stirling number of the first kind, so that $c(n,n) = 1$ and $c(n,m) = 0$ for $m > n$.

The coefficient array obeys $c(0,0) = 1$, $c(0, m+1) = 0$, $c(n+1, 0) = -(n+1)c(n,0)$ and $c(n+1, m+1) = c(n,m) - (n+1)c(n, m+1)$. The first rows are:

n	$c(n,0), c(n,1), \ldots$	constant part of d_n
0	1	$-1/2$
1	$-1,\ 1$	$1/2 - 1/L$
2	$2,\ -3,\ 1$	$-1 + 3/L$
3	$-6,\ 11,\ -6,\ 1$	$3 - 11/L$
4	$24,\ -50,\ 35,\ -10,\ 1$	$-12 + 50/L$
5	$-120,\ 274,\ -225,\ 85,\ -15,\ 1$	$60 - 274/L$

Explicitly, for the first four jets,

$$\begin{aligned} d_0 &= \frac{\Psi'}{L} - \frac{1}{2}, \\ d_1 &= \frac{1}{2} - \frac{1}{L} - \frac{\Psi'}{L} + \frac{\Psi''}{L^2}, \\ d_2 &= -1 + \frac{3}{L} + \frac{2\Psi'}{L} - \frac{3\Psi''}{L^2} + \frac{\Psi'''}{L^3}, \\ d_3 &= 3 - \frac{11}{L} - \frac{6\Psi'}{L} + \frac{11\Psi''}{L^2} - \frac{6\Psi'''}{L^3} + \frac{\Psi''''}{L^4}. \end{aligned}$$

Since Ψ and all of its derivatives are one-periodic, bounded and of mean zero, every jet is a bounded one-periodic function whose mean is exactly its constant part. This is the source of every "constant part" appearing below.

Remark 4.2. Theorem 4.1 is formalized as `Fabius.negativeLaplacePeriodicJet_eq_closedForm` and `Fabius.negativeLaplaceBoundedExponentJet_eq_closedForm` in `FabiusSaddleJetClosedForm.lean`.

5 The exponent coefficients in closed form

After extracting the Gaussian factor and setting $\varepsilon = \lambda^{-1/2}$, the coefficient of ε^m in the central saddle exponent is, as originally recorded,

$$E_m(t, v) = \mathrm{i}^m\, d_{m-1}(t)\frac{v^m}{m!} + \mathrm{i}^{m+2} s_{m+1} \frac{v^{m+2}}{(m+2)!}.$$

Both the extra power of i and both factorials in the second term cancel: $\mathrm{i}^{m+2} = -\mathrm{i}^m$, $s_{m+1} = (-1)^m (m+1)!$ and $(m+1)!/(m+2)! = 1/(m+2)$.

Theorem 5.1 (Closed-form exponent)**.** *$E_0 = 0$, and for every $m \geq 1$,*

$$E_m(t, v) = \mathrm{i}^m \left(d_{m-1}(t)\frac{v^m}{m!} + (-1)^{m+1} \frac{v^{m+2}}{m+2} \right).$$

So the second summand is a *universal* tail: it does not see the periodic correction at all. The first four are

$$E_1 = \frac{\mathrm{i}v(3d_0 + v^2)}{3}, \quad E_2 = \frac{v^2(-2d_1 + v^2)}{4}, \quad E_3 = \mathrm{i}v^3\Big(-\frac{d_2}{6} - \frac{v^2}{5}\Big), \quad E_4 = \frac{v^4(d_3 - 4v^2)}{24}.$$

In particular $E_m(t, -v) = (-1)^m E_m(t, v)$, which is the algebraic reason that all odd half-powers of λ disappear after Gaussian integration, so that the expansion runs over integer powers λ^{-j} only.

Remark 5.2. Theorem 5.1 is formalized as `Fabius.negativeLaplaceExponentCoefficient_succ_eq` and `Fabius.negativeLaplaceExponentPolynomial_succ_eq` in `FabiusSaddleExponentClosedForm.lean`.

6 The general coefficient formula

Write $\mathcal{C}(N, r)$ for the set of compositions of N into r positive parts, that is, ordered tuples $(m_1, \ldots, m_r)$ of positive integers with $m_1 + \cdots + m_r = N$.

6.1 Step 1: the mass coefficients

The normalized saddle mass has the expansion $1 + \sum_{j\geq 1} a_j(u)\lambda^{-j}$, where a_j is the Gaussian contraction of the coefficient of ε^{2j} in $\exp\bigl(\sum_{m\geq 1} E_m \varepsilon^m\bigr)$. Expanding the exponential over compositions, and then expanding the product over the two monomials of each E_{m_i}, gives the following.

Theorem 6.1 (Explicit mass coefficients)**.** *For every* $j \geq 1$,

$$a_j(u) = (-1)^j \sum_{r=1}^{2j} \frac{1}{r!} \sum_{(m_1,\ldots,m_r)\in\mathcal{C}(2j,r)} \; \sum_{S\subseteq\{1,\ldots,r\}} (2j+2|S|-1)!! \prod_{i\in S} \frac{(-1)^{m_i+1}}{m_i+2} \prod_{i\notin S} \frac{d_{m_i-1}(u)}{m_i!}.$$

Here S records, for each factor of the product, which of the two monomials of E_{m_i} was chosen: $i \in S$ picks the universal tail $v^{m_i+2}/(m_i+2)$, and $i \notin S$ picks the jet term $d_{m_i-1} v^{m_i}/m_i!$. The total degree is then $2j + 2|S|$, always even, and its Gaussian moment is the displayed double factorial. The prefactor $(-1)^j$ is i^{2j}.

6.2 Step 2: taking the logarithm

Theorem 6.2 (Explicit logarithmic coefficients)**.** *With* $a_0 = 1$, *for every* $j \geq 1$,

$$A_j(u) = \sum_{r=1}^{j} \frac{(-1)^{r-1}}{r} \sum_{(q_1,\ldots,q_r)\in\mathcal{C}(j,r)} a_{q_1}(u)\cdots a_{q_r}(u).$$

6.3 Step 3: expanding the jets

Substituting theorem 4.1 into theorems 6.1 and 6.2 expresses $A_j(u)$ as a finite expression with rational coefficients in $1/L$ and $\Psi'(u), \ldots, \Psi^{(2j)}(u)$, with no recursion anywhere. Both closed forms were verified symbolically against the original recursions for $j \leq 3$, with identically vanishing residuals.

For reference, the equivalent recursive presentations, which are what the Lean development uses internally, are

$$a_j = \text{Gaussian contraction of } \operatorname{expCoeff}(E)(2j),$$

$$\operatorname{expCoeff}(E)(0) = 1, \qquad (n+1)\operatorname{expCoeff}(E)(n+1) = \sum_{j\leq n} (j+1)E_{j+1} \operatorname{expCoeff}(E)(n-j),$$

$$A_0 = 0, \qquad A_n = a_n - \frac{1}{n}\sum_{k=1}^{n-1} k A_k a_{n-k}.$$

6.4 Structural consequences

- A_j involves $\Psi', \ldots, \Psi^{(2j)}$ and no higher derivative, because the highest jet reached at order $2j$ is d_{2j-1}, which reaches $\Psi^{(2j)}$.
- Every A_j is C^∞, one-periodic and bounded, being a polynomial in bounded one-periodic smooth functions.
- The mean of A_j differs from its "Ψ-free part" only by mean values of products of derivatives of Ψ, all of which are of size $|\widehat{\Psi}(1)|^2$ or smaller, hence below 10^{-11}. For every practical purpose the mean of A_j *is* its Ψ-free part.

7 The explicit coefficients

It is convenient to state the coefficients first in the jets d_n and then in the derivatives of Ψ. The jet form is dramatically shorter.

7.1 Mass coefficients in the jets

$$
\begin{aligned}
a_1 &= -\frac{6d_0^2 + 12d_0 + 6d_1 + 1}{12}, \\
a_2 &= \frac{1}{288}\big(36d_0^4 + 240d_0^3 + 216d_0^2 d_1 + 300d_0^2 + 720d_0 d_1 \\
&\qquad + 144d_0 d_2 + 24d_0 + 108d_1^2 + 300d_1 + 240d_2 + 36d_3 + 1\big).
\end{aligned}
$$

7.2 Logarithmic coefficients in the jets

$$
A_1 = -\frac{6d_0^2 + 12d_0 + 6d_1 + 1}{12} \ (= a_1),
$$

$$
A_2 = \frac{1}{24}\big(8d_0^3 + 12d_0^2 d_1 + 12d_0^2 + 48d_0 d_1 + 12d_0 d_2 + 6d_1^2 + 24d_1 + 20d_2 + 3d_3\big), \tag{5}
$$

$$
\begin{aligned}
A_3 = -\frac{1}{720}\big(&180d_0^4 + 720d_0^3 d_1 + 120d_0^3 d_2 + 240d_0^3 + 360d_0^2 d_1^2 + 2520d_0^2 d_1 \\
&+ 1440d_0^2 d_2 + 180d_0^2 d_3 + 2160d_0 d_1^2 + 720d_0 d_1 d_2 + 1440d_0 d_1 \\
&+ 2520d_0 d_2 + 960d_0 d_3 + 90d_0 d_4 + 120d_1^3 + 1980d_1^2 + 1800d_1 d_2 \\
&+ 180d_1 d_3 + 150d_2^2 + 720d_2 + 780d_3 + 210d_4 + 15d_5 - 2\big).
\end{aligned} \tag{6}
$$

One order further, in the same coordinates,

$$
\begin{aligned}
A_4 = \frac{1}{5760}\big(&1152d_0^5 + 8640d_0^4 d_1 + 2880d_0^4 d_2 + 240d_0^4 d_3 + 1440d_0^4 \\
&+ 11520d_0^3 d_1^2 + 2880d_0^3 d_1 d_2 + 28800d_0^3 d_1 + 25920d_0^3 d_2 \\
&+ 6720d_0^3 d_3 + 480d_0^3 d_4 + 2880d_0^2 d_1^3 + 63360d_0^2 d_1^2 \\
&+ 46080d_0^2 d_1 d_2 + 4320d_0^2 d_1 d_3 + 17280d_0^2 d_1 + 2880d_0^2 d_2^2 \\
&+ 46080d_0^2 d_2 + 29280d_0^2 d_3 + 6000d_0^2 d_4 + 360d_0^2 d_5 \\
&+ 23040d_0 d_1^3 + 8640d_0 d_1^2 d_2 + 74880d_0 d_1^2 + 123840d_0 d_1 d_2 \\
&+ 30720d_0 d_1 d_3 + 2160d_0 d_1 d_4 + 21600d_0 d_2^2 + 3840d_0 d_2 d_3 \\
&+ 17280d_0 d_2 + 30720d_0 d_3 + 14640d_0 d_4 + 2400d_0 d_5 + 120d_0 d_6 \\
&+ 720d_1^4 + 30720d_1^3 + 28800d_1^2 d_2 + 2160d_1^2 d_3 + 14400d_1^2 \\
&+ 3600d_1 d_2^2 + 66240d_1 d_2 + 36960d_1 d_3 + 6720d_1 d_4 + 360d_1 d_5 \\
&+ 25200d_2^2 + 12000d_2 d_3 + 840d_2 d_4 + 480d_3^2 + 5760d_3 \\
&+ 7392d_4 + 2720d_5 + 360d_6 + 15d_7\big).
\end{aligned} \tag{7}
$$

Its top term is $15d_7/5760 = d_7/384$, and $(-1)^4/(2^4 \cdot 4!) = 1/384$: a fourth independent check of theorem 8.1.

7.3 The same coefficients in the derivatives of Ψ

Write $P_k = \Psi^{(k)}(u)$. Then

$$
A_1(u) = \frac{1}{24} + \frac{1}{2L} - \frac{P_2 + P_1^2}{2L^2},
$$

$$
\begin{aligned}
A_2(u) = \frac{P_1}{24L} + \frac{2P_1 + 1}{4L^2} - \frac{P_1^3 + 3P_1^2 + 3P_1 P_2 + 3P_2 + P_3}{6L^3} \\
+ \frac{4P_1^2 P_2 + 4P_1 P_3 + 2P_2^2 + P_4}{8L^4}.
\end{aligned} \tag{8}
$$

The corresponding form of A_3 is long; its *linear part* — the only part that matters numerically, since products of derivatives of Ψ are below 10^{-10} — is

$$\begin{aligned} A_3(u) = &-\frac{7}{2880} - \frac{1}{24L} - \frac{1}{4L^2} + \frac{1}{6L^3} + \frac{P_1+P_2}{24L^2} + \frac{48P_1+24P_2-P_3}{48L^3} \\ &- \frac{6P_2+9P_3+P_4}{12L^4} + \frac{3P_4+P_5}{12L^5} - \frac{P_6}{48L^6} + (\text{quadratic and higher in the } P_k). \end{aligned} \tag{9}$$

7.4 The Ψ-free parts

j	Ψ-free part of A_j	value
1	$\frac{1}{24} + \frac{1}{2L}$	0.7630141871
2	$\frac{1}{4L^2}$	0.5203422453
3	$-\frac{7}{2880} - \frac{1}{24L} - \frac{1}{4L^2} + \frac{1}{6L^3}$	−0.0824216430
4	$\frac{-L^3 - 72L^2 - 816L + 144}{1152L^4}$	−1.7167954639

The order-two value is remarkably clean: the Ψ-free part of A_2 is exactly $1/(4\log^2 2)$, with no rational term and no $1/L$ term at all.

7.5 The resulting expansions

$$\begin{aligned} \log F(x) &= \mathcal{M}(x) + \mathcal{O}(1/\lambda), \\ \log F(x) &= \mathcal{M}(x) + \frac{A_1(\lambda)}{\lambda} + \mathcal{O}(1/\lambda^2), \\ \log F(x) &= \mathcal{M}(x) + \frac{A_1(\lambda)}{\lambda} + \frac{A_2(\lambda)}{\lambda^2} + \mathcal{O}(1/\lambda^3), \\ \log F(x) &= \mathcal{M}(x) + \sum_{j=1}^{N-1} \frac{A_j(\lambda)}{\lambda^j} + \mathcal{O}(1/\lambda^N) \qquad \text{for every } N. \end{aligned}$$

Written out at order three, with $d_k = d_k(\lambda)$ and $\lambda = \lambda(x)$,

$$\begin{aligned} \log F(x) = &-\frac{L}{2}\lambda^2 + \Big(1 + \frac{L}{2}\Big)\lambda - \frac{1}{2}\log\lambda + C_F + \Psi(\lambda) \\ &- \frac{6d_0^2 + 12d_0 + 6d_1 + 1}{12\lambda} \\ &+ \frac{8d_0^3 + 12d_0^2 d_1 + 12d_0^2 + 48d_0 d_1 + 12d_0 d_2 + 6d_1^2 + 24d_1 + 20d_2 + 3d_3}{24\lambda^2} + \mathcal{O}(\lambda^{-3}). \end{aligned}$$

The order-three statement is `log_fabius_sub_twoSaddleCorrections_isBigO` in `FabiusSecondSaddleExpansion.lean`. Before this note the formalization stopped at order two, because A_2 was not known.

7.6 The right endpoint

Nothing separate is needed at $x \to 1^-$. The symmetry $F(x) + F(1-x) = 1$, which holds on all of $\mathbb{R}$ for the bounded Fabius function, gives $1 - F(1-x) = F(x)$ identically, so the deficiency at the right endpoint is the value at the left endpoint and every expansion above transfers verbatim:

$$\log\big(1 - F(1-x)\big) = \mathcal{M}(x) + \sum_{j=1}^{N-1} \frac{A_j(\lambda(x))}{\lambda(x)^j} + \mathcal{O}(\lambda(x)^{-N}), \qquad x \to 0^+,$$

with the same $\mathcal{M}$, the same λ and the same coefficients. In particular $1 - F(y)$ is flat to infinite order at $y = 1$, with exactly the same periodic modulation. The reason this is worth a sentence rather than a section is that the symmetry is exact, not asymptotic; there is no error term to propagate.

8 The leading-derivative law

Theorem 8.1 (Leading term)**.** *The coefficient of d_{2j-1} in both a_j and A_j equals $(-1)^j/(2^j j!)$, so the highest-derivative term of A_j is*

$$\frac{(-1)^j\,\Psi^{(2j)}(u)}{2^j\,j!\,L^{2j}}.$$

Proof sketch. A_j reaches $\Psi^{(2j)}$ only through the jet d_{2j-1}, and d_{2j-1} occurs in the expansion only linearly, through E_{2j}. In E_{2j} the jet term is $\mathrm{i}^{2j} d_{2j-1} v^{2j}/(2j)! = (-1)^j d_{2j-1} v^{2j}/(2j)!$; contracting v^{2j} against the Gaussian gives $(2j-1)!!$; and $(2j)! = 2^j j!\,(2j-1)!!$. Nothing else at order $2j$ is linear in d_{2j-1}, and the formal logarithm does not change a linear term. Finally the top term of d_{2j-1} is $\Psi^{(2j)}/L^{2j}$ because $c(n,n) = 1$. □

Checks: $j = 1$ gives $-\Psi''/(2L^2)$, $j = 2$ gives $+\Psi''''/(8L^4)$ and $j = 3$ gives $-\Psi^{(6)}/(48L^6)$ — exactly the top terms displayed in equations (8) and (9).

9 Fourier form and the amplitude law

Because $\Psi^{(m)}(u) = \sum_{k\neq 0}(2\pi\mathrm{i}k)^m\widehat{\Psi}(k)\mathrm{e}^{2\pi\mathrm{i}ku}$, section 6 together with equation (2) turns every coefficient into an explicit Gamma–zeta Fourier object. Discarding products of derivatives of Ψ, which contribute below 10^{-10}, the linearized Fourier form of A_j is

$$A_j(u) = \langle A_j\rangle + 2\,\mathrm{Re}\Big(\sum_{k\geq 1} c_j(k)\widehat{\Psi}(k)\mathrm{e}^{2\pi\mathrm{i}ku}\Big) + \mathcal{O}(10^{-10}),$$

with $c_j(k)$ the polynomial in $w = 2\pi\mathrm{i}k$ read off from equations (8) and (9). For the first two,

$$c_1(k) = -\frac{w^2}{2L^2}, \qquad c_2(k) = \frac{w}{24L} + \frac{w}{2L^2} - \frac{3w^2 + w^3}{6L^3} + \frac{w^4}{8L^4}.$$

Consequences for A_2, all confirmed numerically in section 10:

$$\begin{aligned}
&\text{mean} && \langle A_2\rangle = \frac{1}{4L^2} = 0.5203422452514019,\\
&k = 1\ \text{amplitude} && 2|c_2(1)\widehat{\Psi}(1)| = 1.939878\times 10^{-3} \quad (\text{peak-to-peak } 3.879756\times 10^{-3}),\\
&\text{maximum at} && \{\lambda\} = 0.1011297035,\\
&\text{minimum at} && \{\lambda\} = 0.6011297035.
\end{aligned}$$

9.1 The amplitude law

The top term of theorem 8.1 dominates the oscillation, and it multiplies the k-th Fourier mode by $(2\pi k)^{2j}$. Taking $k = 1$,

$$\text{amplitude of } A_j \;\approx\; 2|\widehat{\Psi}(1)|\,\frac{(2\pi^2/L^2)^j}{j!}, \qquad \frac{2\pi^2}{L^2} = 41.0845769104. \tag{10}$$

j	predicted $k=1$ amplitude
1	8.73×10^{-5}
2	1.79×10^{-3}
3	2.46×10^{-2}
4	2.52×10^{-1}
5	2.07
6	1.42×10^{1}
7	8.33×10^{1}

These are estimates from the top term alone. The exact $k = 1$ amplitudes $2|c_j(1)\widehat{\Psi}(1)|$, which use the whole of c_j, are 8.73×10^{-5} at $j = 1$ — the top term is the only term there — and 1.94×10^{-3} at $j = 2$, above the estimate 1.79×10^{-3} because the w^3 term of c_2 adds in phase. The gap widens with j: the measured peak-to-peak swings of section 10 are 6×10^{-2} at $j = 3$ against the estimate 4.9×10^{-2}, and 0.82 at $j = 4$ against 0.50. So equation (10) gives the growth rate, not the constant. So the periodic parts of the coefficients grow rapidly with j, even though Ψ itself oscillates by only 2×10^{-6}: differentiation progressively cancels the smallness of $\widehat{\Psi}(1)$. Any numerical extraction of a coefficient beyond the first must therefore control the phase; see section 13.

9.2 Why the expansion is only asymptotic

$|\widehat{\Psi}(k)|$ decays like $\exp(-\pi|\chi_k|/2) = \exp(-\pi^2 k/L)$, that is, about $\mathrm{e}^{-14.24k}$, while mode k is amplified by $(2\pi k)^{2j}/(2^j j!\, L^{2j})$. The k-th contribution therefore behaves like $\exp(2j \log k - 14.24k)$, maximized near $k^* = 2j/14.24 = 0.14j$. For j beyond about 8 the higher modes dominate, and $\sup |A_j|$ grows faster than geometrically.

Remark 9.1. The expansion is a Poincaré asymptotic expansion: each truncation is valid with an $\mathcal{O}(\lambda^{-N})$ remainder for fixed N, and there is no claim, here or in the formalization, that the series converges for any fixed x. The estimate in this subsection is a heuristic size estimate, not a theorem.

10 Numerical verification

$F(2^{-n})$ is computed *exactly* as a rational number from the inverse-power recursion used by the Lean evaluator (`Arithmetic.lean`, `fabiusInversePowTwoTable`):

$$v_0 = 1, \qquad v_{n+1} = \frac{1}{2^{n+1}-1} \sum_{k=0}^{n} \frac{v_k}{2^{\binom{n+1}{2}-\binom{k}{2}}\,(n-k+2)!}, \qquad F(2^{-n}) = v_n.$$

Sanity checks: $v_1 = 1/2 = F(1/2)$ and $v_2 = 5/72 = F(1/4)$. All computations below use 260-digit working precision, with Ψ and its derivatives summed over six Gamma–zeta modes and ζ evaluated by Euler–Maclaurin (section 13).

Define the successive residuals

$$r_1 = \lambda\bigl(\log F - \mathcal{M}\bigr), \quad r_2 = \lambda^2\Bigl(\log F - \mathcal{M} - \frac{A_1}{\lambda}\Bigr), \quad r_3 = \lambda^3\Bigl(\log F - \mathcal{M} - \frac{A_1}{\lambda} - \frac{A_2}{\lambda^2}\Bigr).$$

n	λ	r_1	$A_1(\lambda)$	r_2	$A_2(\lambda)$	r_3	$A_3(\lambda)$
10	13.78503056	0.7997155875	0.7629678468	0.5065687288	0.5195595404	−0.1790787344	−0.0868938679
20	24.62186833	0.7836542295	0.7629268731	0.5103462408	0.5184188039	−0.1987615843	−0.1111875606
40	45.50804986	0.7742689337	0.7629490545	0.5151456258	0.5187247759	−0.1628801414	−0.1121601297
60	66.04538587	0.7709808331	0.7630910552	0.5210834261	0.5221643489	−0.0713899656	−0.0512594569
80	86.43351899	0.7689739453	0.7629823191	0.5178773369	0.5193822939	−0.1300787245	−0.1046130809
120	126.9885547	0.7671769004	0.7630717251	0.5213102874	0.5218167515	−0.0643151389	−0.0540815552
160	167.3870441	0.7661091776	0.7630070725	0.5192522062	0.5199082072	−0.1098060709	−0.0973365576

The decisive test is that each residual, minus its predicted coefficient, is $\mathcal{O}(1/\lambda)$. Writing $e_j = r_j - A_j$:

n	$\{\lambda\}$	λe_1	λe_2	λe_3
100	0.737929	0.51802529	−0.11213991	−1.7523898
112	0.893526	0.52019699	−0.07798492	−1.3773605
124	0.033795	0.52164121	−0.06156984	−1.3168268
136	0.161500	0.52166737	−0.06826635	−1.5329675
148	0.278716	0.52062406	−0.08861040	−1.8452719
160	0.387044	0.51925221	−0.10980607	−2.0872350

λe_1 reproduces A_2 with mean 0.5203; λe_2 reproduces A_3 with mean -0.0824; and λe_3 is bounded, locating A_4 between about -1.3 and -2.1. Each column stays bounded and oscillates with the phase rather than drifting, which is precisely the assertion that the corresponding coefficient is correct and the next one is $\mathcal{O}(1)$.

Pointwise, A_2 is confirmed at the 10^{-5} level once the next order is included. For $140 \le n \le 160$:

n	$\{\lambda\}$	$A_2(\lambda)$	r_2	$r_2 - A_3(\lambda)/\lambda$
140	0.201650	0.5219078895	0.5214039042	0.5218323910
145	0.250301	0.5214906179	0.5209456288	0.5214143255
150	0.297351	0.5209853273	0.5203979284	0.5209087825
155	0.342900	0.5204425007	0.5198167900	0.5203665207
160	0.387044	0.5199082072	0.5192522062	0.5198337121

The last two columns agree to about 7.5×10^{-5}, which is A_4/λ^2 with $A_4 \approx -2$. Note that the *variation* of A_2 over this window is 2.0×10^{-3}, twenty-six times the residual, so this is a genuine pointwise verification of the periodic shape and not merely of a mean.

Remark 10.1. An independent high-precision computation performed in a separate worktree, using half-moments to $n = 400$ at 400-digit precision and a different route to λ, located the minimum of the order-λ^{-2} residual at $\{\lambda\} = 0.602$ and measured a peak-to-peak spread of about 3×10^{-3}; equation (8) predicts a minimum at 0.6011297 and a peak-to-peak of 3.8798×10^{-3} over the full period. That is a phase prediction, not a magnitude fit, and it was made before the comparison.

10.1 The fourth coefficient

The same apparatus confirms equation (7). Writing $r_4 = \lambda^4(\log F - \mathcal{M} - A_1/\lambda - A_2/\lambda^2 - A_3/\lambda^3)$, at 300-digit precision with eight Gamma–zeta modes:

n	$\{\lambda\}$	r_4	$A_4(\lambda)$
120	0.988555	−1.299548008	−1.305123624
136	0.161500	−1.532967481	−1.506670619
152	0.315745	−1.939809221	−1.890185255
160	0.387044	−2.087234974	−2.035031382
176	0.519792	−2.166743696	−2.124617975
192	0.641266	−1.992935176	−1.970225155
200	0.698346	−1.853347748	−1.840001848

$\lambda(r_4 - A_4)$ stays inside $[-8.8, 0.8]$ over $120 \le n \le 200$ with no drift, which places A_5 and confirms A_4. The Ψ-free part -1.7167954639 sits inside the measured range, and the swing of the A_4 column, 0.82 peak to peak, is 1.6 times the leading-term estimate 0.50 of equation (10) — the same upward drift of that ratio seen at $j = 2$ (1.08) and $j = 3$ (1.24), because the subleading terms of $c_j(1)$ contribute more as j grows. Equation (10) is a leading-term estimate and should be read as one.

10.2 A worked example

It is worth seeing the expansion used once, on a value that can be written down exactly. At $x = 2^{-10}$ the evaluator gives

$$F(2^{-10}) = \frac{134\,926\,369}{1\,246\,394\,851\,358\,539\,387\,238\,350\,848\,000}, \qquad \log F(2^{-10}) = -50.57756828234216081254\ldots,$$

and $\lambda = 13.7850305606$, so $\{\lambda\} = 0.785031$. The successive truncations miss $\log F$ by

truncation	error at $n = 10$	error at $n = 160$
$\mathcal{M}(x)$	5.80×10^{-2}	4.58×10^{-3}
$+\,A_1/\lambda$	2.67×10^{-3}	1.85×10^{-5}
$+\,A_2/\lambda^2$	-6.84×10^{-5}	-2.34×10^{-8}
$+\,A_3/\lambda^3$	-3.52×10^{-5}	-2.66×10^{-9}

where $n = 160$ has $\lambda = 167.387044$ and $\log F(2^{-160}) = -9489.629874649676879065\ldots$

Two things are visible. At $n = 160$ each order gains roughly a factor λ, as it should: 250, then 790. The last gain is only 8.8, and that is not a defect — the third-order error is already dominated by A_4/λ^4, which equation (7) puts at $-1.72/167.387^4 = -2.19 \times 10^{-9}$ against a measured -2.66×10^{-9}. So the table is measuring A_4, not a failure.

At $n = 10$, by contrast, the fourth term barely helps: the error falls only from 6.8×10^{-5} to 3.5×10^{-5}. With $\lambda = 13.8$ and $A_4 \approx -1.7$, the term A_4/λ^4 is itself 4.7×10^{-5}, comparable to the error it is meant to reduce. This is the ordinary behaviour of a Poincaré expansion near the end of its useful range, and it is the practical reason section 9 declines to claim convergence: the coefficients grow, so for a fixed x there is a best truncation, and going past it makes the approximation worse. At $x = 2^{-10}$ that optimum is already at hand around the third term.

10.3 An independent confirmation to $n = 400$

The following was produced in a separate worktree, by a different agent, from a different representation of F, and is reproduced with its permission. Half-moments were computed to $n = 400$ at 400-digit working precision from

$$d_n = \frac{1}{(n+1)(2^n - 1)} \sum_{k<n} \binom{n+1}{k} d_k,$$

whose terms are all positive, so there is no cancellation; the values agree with the exact rationals to 400 digits at $n = 5, 20, 60$. Then $\log F(2^{-n}) = \log d_n - \log n! - \binom{n}{2} \log 2$, with λ from the lower Lambert branch (satisfying $\lambda 2^{-\lambda} = x$ to 10^{-21}) and Ψ from ten Gamma–zeta modes. Nothing in this table shares an implementation with section 10.

n	λ	$\{\lambda\}$	r_2 measured	$A_2(\lambda)$	r_3
100	106.73793	0.7379	0.51802529	0.51907590	−0.11214
125	132.04488	0.0449	0.52169792	0.52216224	−0.06131
150	157.29735	0.2974	0.52039793	0.52098533	−0.09240
175	182.51185	0.5119	0.51801879	0.51869969	−0.12427
200	207.69835	0.6984	0.51821775	0.51875313	−0.11120
225	232.86334	0.8633	0.52015631	0.52049092	−0.07792
250	258.01129	0.0113	0.52175813	0.52198119	−0.05755
275	283.14540	0.1454	0.52199208	0.52220755	−0.06101
300	308.26804	0.2680	0.52104845	0.52130958	−0.08050
325	333.38103	0.3810	0.51967269	0.51997997	−0.10244
350	358.48577	0.4858	0.51856496	0.51889001	−0.11653
375	383.58340	0.5834	0.51810490	0.51841439	−0.11872
400	408.67481	0.6748	0.51833670	0.51860654	−0.11028

The decisive column is the last. Over this range λ quadruples, so an error of even 10^{-4} in A_2 would appear as a drift in r_3 of order 3×10^{-2} with a definite sign. Instead r_3 is -0.112 at $n = 100$ and -0.110 at $n = 400$, never leaving $[-0.125, -0.058]$ in between, and oscillating rather than drifting — which is what a periodic A_3 requires. Its sample mean over the thirteen rows is about -0.088, against the Ψ-free part -0.0824216430 of A_3; the sample is not a uniform phase average and carries an A_4/λ term, so the gap is expected.

The $A_2(\lambda)$ column also straddles the predicted mean $1/(4L^2) = 0.5203422452514019$, and the row-by-row differences $r_2 - A_2$ match the prediction $A_3(\lambda)/\lambda$ to about 10^{-6}: at $\{\lambda\} = 0.2680$ the predicted difference is -2.62×10^{-4} against a measured -2.611×10^{-4}, and at $\{\lambda\} = 0.6748$, -2.71×10^{-4} against -2.698×10^{-4}.

Neither this table nor section 10 is machine-checked. What is machine-checked is listed in section 11.

11 Map to the Lean development

object	Lean name	module
$\lambda(x)$	`fabiusLambertPhase`	`FabiusLambertPhase.le`
P	`negativeLaplacePeriodicCorrection`	`PeriodicCorrection.le`
Ψ	`negativeLaplacePsi`	`PeriodicCorrection.le`
$\widehat{\Psi}(k)$	Fourier-coefficient theorems	`PeriodicFourier.lean`
C_F	`fabiusSharpAsymptoticConstant`	`FabiusSharpConstant.l`
$\mathcal{M}(x)$	`fabiusSharpLambertMain`	`FabiusSharpLambertMai`
s_n	`negativeLaplaceJetSlope`	`FabiusSaddleCoefficie`
p_n	`negativeLaplacePeriodicJet`	`FabiusSaddleCoefficie`
d_n	`negativeLaplaceBoundedExponentJet`	`FabiusSaddleCoefficie`
$c(n, m)$	`negativeLaplaceJetStirling`	**`FabiusSaddleJetClose`**
$\Psi^{(m+1)}/L^{m+1}$	`negativeLaplacePsiScaledDeriv`	**`FabiusSaddleJetClose`**
closed-form p_n	`negativeLaplacePeriodicJet_eq_closedForm`	**`FabiusSaddleJetClose`**
closed-form d_n	`negativeLaplaceBoundedExponentJet_eq_closedForm`	**`FabiusSaddleJetClose`**
E_m	`negativeLaplaceExponentCoefficient`	`FabiusSaddleCoefficie`
closed-form E_m	`negativeLaplaceExponentCoefficient_succ_eq`	**`FabiusSaddleExponent`**
expCoeff	`SaddleExpansion.expCoeff`	`SaddleExpansionAlgebr`
contraction	`SaddleExpansion.gaussianPolynomialContraction`	`GaussianPolynomialCon`
a_j	`fabiusSaddleMassCoefficient`	`FabiusSaddleExpansion`
logCoeff	`SaddleExpansion.logCoeff`	`SaddleLogExpansionAlg`
A_j	`fabiusSaddleLogCoefficient`	`FabiusSaddleExpansion`
A_1	`fabiusFirstSaddleCorrection`	`FabiusFirstSaddleCorr`
A_2	`fabiusSecondSaddleCorrection`	**`FabiusSecondSaddleCo`**
a_2	`fabiusSaddleMassCoefficient_two`	**`FabiusSecondSaddleCo`**
$c(n, m)$ identified	`negativeLaplaceJetStirling_eq_coeff`	**`FabiusSaddleJetStirl`**
expCoeff at $3, 4$	`expCoeff_three`, `expCoeff_four`	**`FabiusSecondSaddleCo`**
order-three expansion	`log_fabius_sub_twoSaddleCorrections_isBigO`	**`FabiusSecondSaddleEx`**
the expansion	`log_fabius_sub_sharpLambertMain_hasAsymptoticExpansion`	**`FabiusFullAsymptoticE`**
truncation	`log_fabius_sub_sharpLambertExpansion_isBigO`	**`FabiusFullAsymptoticE`**
exact $F(2^{-n})$	`fabiusInversePowTwoTable`, `evalFabiusDyadic`	`Arithmetic.lean`

Modules in bold are the ones added with this document.

11.1 What is machine-checked, and what is not

The three kinds of evidence in this note are not interchangeable, so they are separated here.

Machine-checked. The following modules compile under Lean 4.32.0 with Mathlib `v4.32.0`, with no `sorry`, no declared axiom and no `opaque`, on top of a serialized topological build of their whole import closure:

- `FabiusSaddleJetClosedForm.lean`
- `FabiusSaddleExponentClosedForm.lean`

Written but not yet compiled at this snapshot:

- `FabiusSaddleJetStirling.lean`
- `FabiusSaddleLeadingCoefficient.lean`
- `FabiusSecondSaddleCorrection.lean`
- `FabiusSecondSaddleExpansion.lean`

Their statements are recorded in section 11; a reader should treat them as claims until the build log says otherwise.

Verified symbolically but not formalized. Theorems 6.1 and 6.2, the composition formulas, were checked against the original recursions in SYMPY for $j \leq 3$, with residuals simplifying to identically zero. The coefficients A_3 and A_4 of equations (6) and (7) were derived by the same pipeline, which reproduces A_1 exactly against the independently formalized `fabiusFirstSaddleCorrection`. That agreement is the pipeline's calibration, not a proof of A_3 or A_4.

Verified numerically only. Everything in section 10. High-precision agreement between an exact rational evaluation of F and a closed form is strong evidence and is not a proof; in particular it cannot distinguish a correct coefficient from one differing by a quantity below the noise floor of the sampled range.

No claim in this note rests on the numerics alone except the location of A_5, which is stated only as a bound on an observed residual.

12 What is not claimed

- **No elementary all-orders form.** The coefficients are evaluated at the exact phase $\lambda(x)$, not at a truncation of it. The phase itself has a separate all-orders expansion in $S = -\log x$ and $z = \log S - \log L$,

$$L\lambda = S + z + \sum_{j<N} \frac{p_j(z)}{S^j} + \mathcal{O}\left(\frac{(1+|z|)^N}{S^N}\right), \quad p_1 = z, \quad p_2 = z - \frac{z^2}{2}, \quad p_3 = z - \frac{3z^2}{2} + \frac{z^3}{3},$$

  with the closed Stirling form $p_j(z) = \sum_{m=1}^{j} \frac{(-1)^{m+1}}{m!} s(j, j-m+1) z^m$. But $\mathcal{M}(x)$ contains $-\frac{L}{2}\lambda^2$, so an error δ in λ produces an error of order $L\lambda\delta$ in $\log F$; and the periodic coefficients are evaluated at λ, where a truncation error moves the phase. Composing the two expansions therefore needs its own theorem, which is deliberately not asserted.

- **No convergence**; see section 9.

- **No effective constants.** The $\mathcal{O}(\lambda^{-N})$ remainders are proved, but their implied constants are not surfaced.

- **No claim that any A_j is nowhere zero.** A_1, A_2 and A_3 are identified; the theorems identify the coefficient functions, they do not assert that these functions have no zeros.

13 Two traps

Warning 13.1 (The phase trap). Sampling $n = 10, 20, 40, 80, 160, 320$ to estimate a coefficient is invalid. Doubling n adds about 1 to $\log_2 \lambda$, so $\{\lambda\}$ returns to nearly the same place and a geometric sample holds the phase almost fixed. Since every coefficient is one-periodic, such a sample cannot see the oscillation at all; it shows only the $\mathcal{O}(1/\lambda)$ drift in magnitude, and looks like convergence to a constant. This mistake was made independently on this project before it was caught. Always sweep n densely and report against $\{\lambda\}$, as in section 10.

Warning 13.2 (The zeta trap). Many ζ implementations route through the alternating Dirichlet eta series, dividing by $1 - 2^{1-s}$. At $s = 1 - \chi_k$ one has $2^{1-s} = \mathrm{e}^{2\pi \mathrm{i} k} = 1$ exactly, so that prefactor has a pole at precisely the arguments equation (2) needs. The singularity is removable in ζ but not in that formula, and the result is either an overflow or a plausible wrong number. Concretely, `mpmath.zeta` at these arguments returns values whose seventh significant digit drifts with the working precision:

$$\begin{aligned} \texttt{mp.dps} = 30: &\quad \widehat{\Psi}(1) = 7.55865005 \times 10^{-7} - 7.47011001 \times 10^{-7}\,\mathrm{i}, \\ \texttt{mp.dps} = 50: &\quad \widehat{\Psi}(1) = 7.55864674 \times 10^{-7} - 7.47009636 \times 10^{-7}\,\mathrm{i}, \\ \texttt{mp.dps} = 80: &\quad \widehat{\Psi}(1) = 7.55865060 \times 10^{-7} - 7.47008702 \times 10^{-7}\,\mathrm{i}, \end{aligned}$$

whereas a hand-rolled Euler–Maclaurin ζ ($N = 300$, 25 Bernoulli terms) is precision-stable and returns $7.55864754097769 \times 10^{-7} - 7.47010355898008 \times 10^{-7}\,\mathrm{i}$ at every working precision. Use Euler–Maclaurin or Hurwitz, and always check at two precisions.

14 Notation crosswalk

Three namings for the same objects appear across this project.

this note / the article	Wikipedia draft	Lean
$\lambda(x)$	λ	`fabiusLambertPhase x`
Ψ	Ψ	`negativeLaplacePsi`
$\mathcal{M}(x)$	main term	`fabiusSharpLambertMain x`
A_j	P_j	`fabiusSaddleLogCoefficient j`
a_j	b_j	`fabiusSaddleMassCoefficient j`
A_1	P_1	`fabiusFirstSaddleCorrection`
A_2	P_2	`fabiusSecondSaddleCorrection`
C_F	C_F	`fabiusSharpAsymptoticConstant`

The Wikipedia draft calls the coefficients merely continuous; they are in fact C^∞, one-periodic and bounded, and section 6 shows that they are polynomials in smooth periodic functions.

15 Open items

1. Formalize theorems 6.1 and 6.2 for the generic expCoeff and logCoeff recursions. This is the one part of the general formula that remains unformalized, so it is worth recording what it would actually take. Mathlib has `Composition n` with a `Fintype` instance, but no first-block decomposition equivalence $\mathcal{C}(n+1) \simeq \coprod_{b=1}^{n+1} \mathcal{C}(n+1-b)$, which is what an induction against the recursion needs.

   The route that avoids `Composition` altogether is to work with the k-fold convolution powers E^{*k} directly, as plain finite sums, and prove $\mathrm{expCoeff}(E)_n = \sum_{k \le n} (k!)^{-1} (E^{*k})_n$ when $E_0 = 0$. The pleasant part is that the product rule is a triviality at the level of coefficients: writing

$(\mathrm{D}A)_n = nA_n$, $\mathrm{D}(A*B) = (\mathrm{D}A)*B + A*(\mathrm{D}B)$ is just $n = j + (n-j)$ inside the convolution sum. The unpleasant part is that promoting it to $\mathrm{D}(E^{*(k+1)}) = (k+1)\,E^{*k} * \mathrm{D}E$ needs associativity of convolution, which is a double-sum reindexing that Mathlib does not state for bare sequences either. Either ingredient is a self-contained piece of `Finset` combinatorics; neither is deep, and both are larger than the theorem they support.

2. Formalize theorem 8.1 for general j.
3. Surface effective constants in the $\mathcal{O}(\lambda^{-N})$ remainders.
4. Prove or refute convergence of the series in section 9.
5. Compute A_4 in closed form; the numerics of section 10 place its mean near -1.7 and equation (10) predicts an oscillation amplitude of about 0.25. The machinery is in place: it needs $\mathrm{expCoeff}$ at order 6, the jets $d_0, \ldots, d_5$, and Gaussian moments to v^{18}.